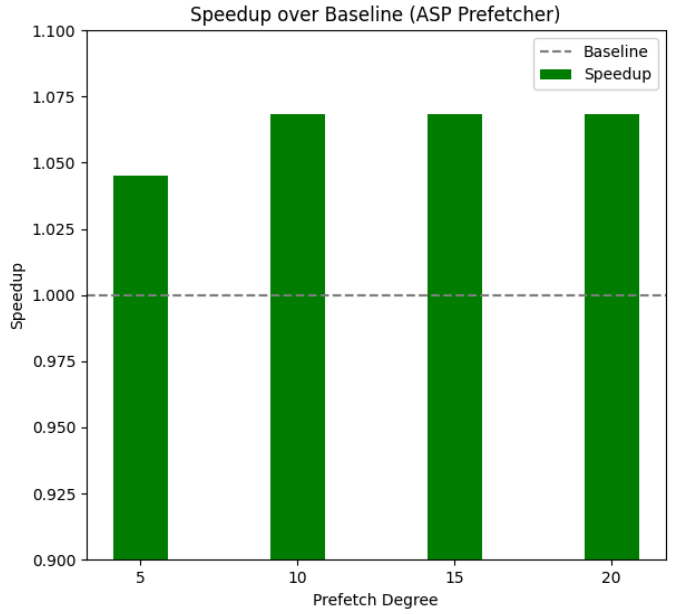
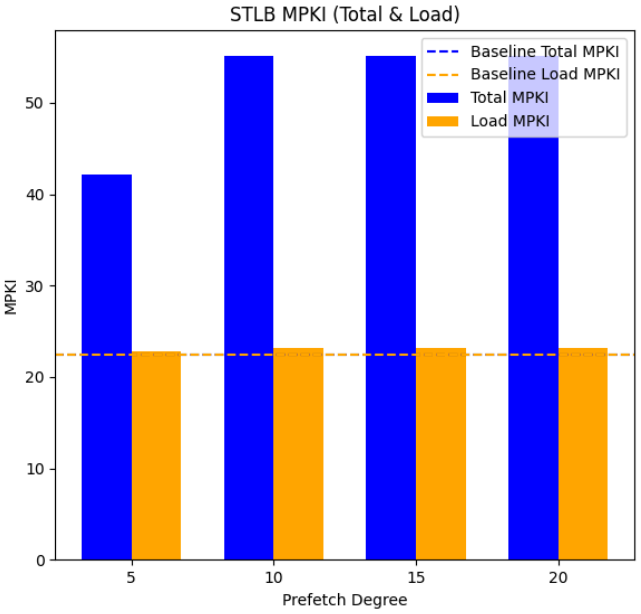
Programming Assignment 2 - Lehra Do! Prefetchers
CS 683: Advanced Computer Architecture, Autumn 2024

Team - Prefetchers Data Le Jayenge
Mriganko Shekhar Dutta 24M2134
Digvijay Shalikrao Patil 24M2105

Task 1: TLB Prefetcher (5 points)

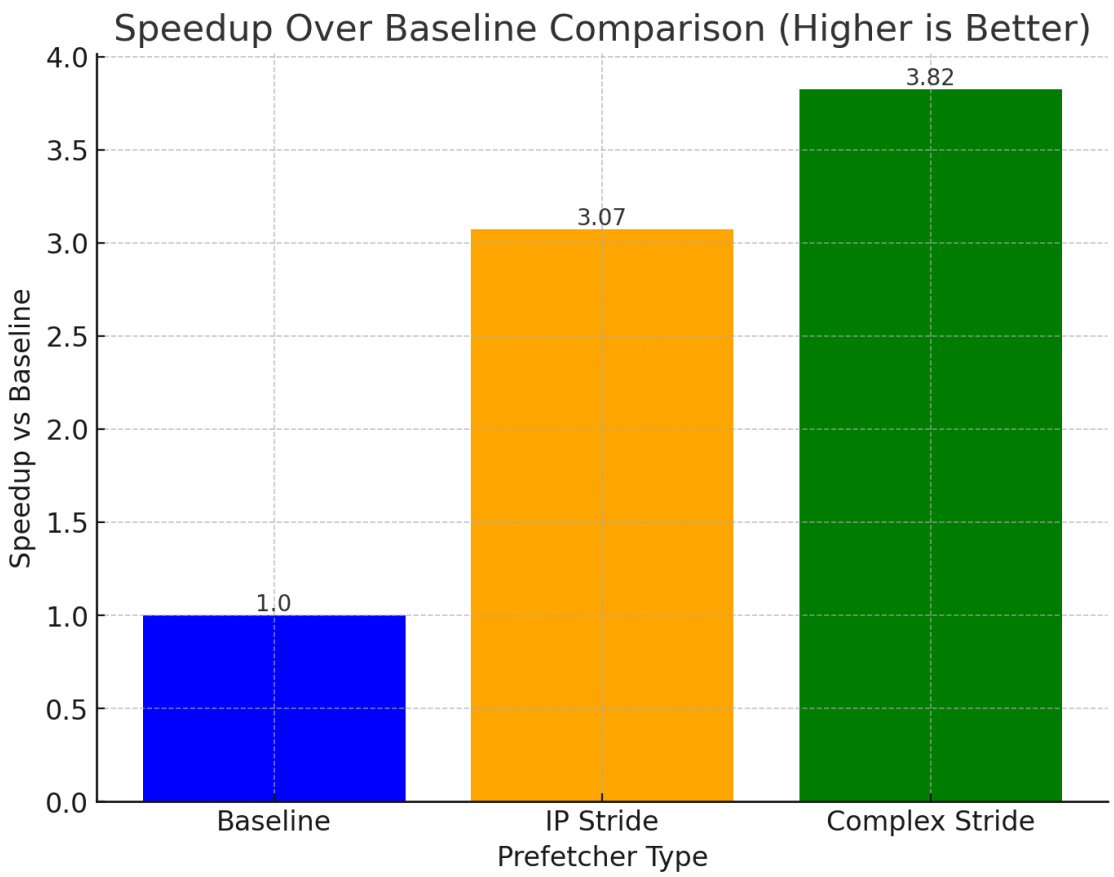
Configuration	Degree	IPC	Seedup over Baseline	Stlb Total MPKI	Stlb load translation MPKI
Baseline	-	0.282799	1.00	22.4811	22.4811
ASP	5	0.295571	1.0452	42.1733	22.7767
ASP	10	0.302137	1.0686	55.1802	23.1197
ASP	15	0.302137	1.0686	55.1802	23.1197
ASP	20	0.302137	1.0686	55.1802	23.1197

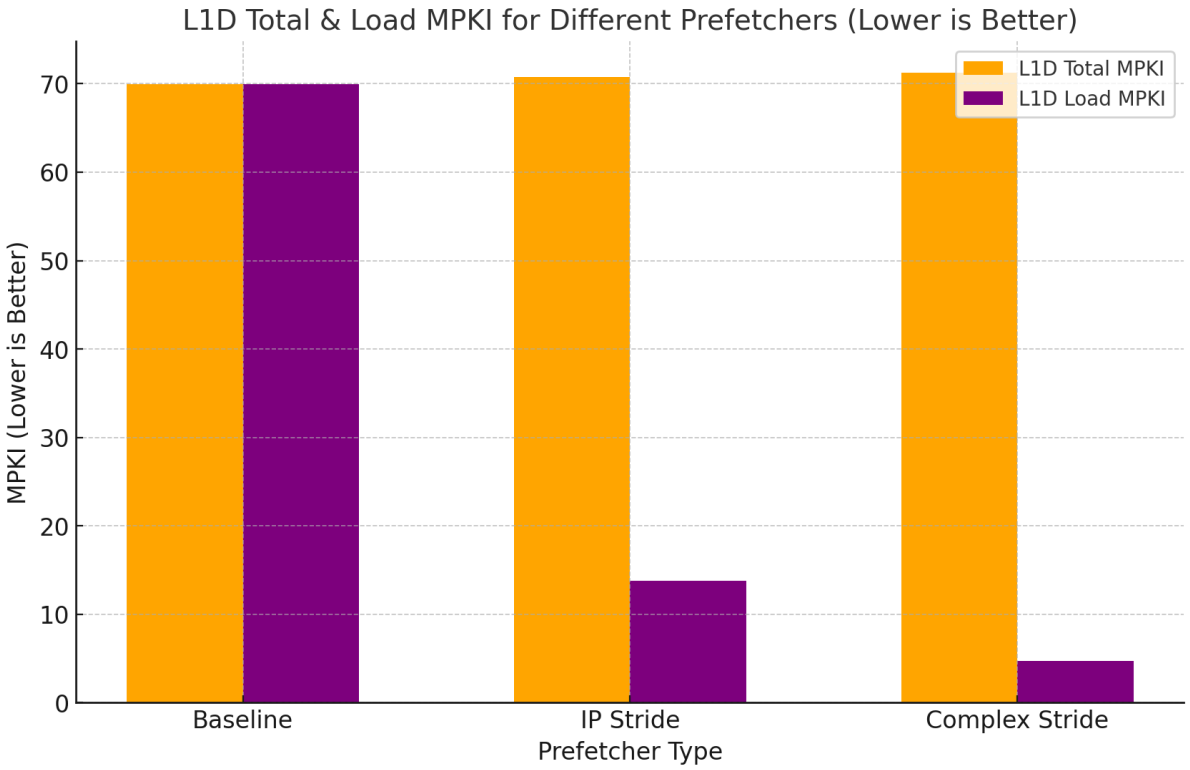




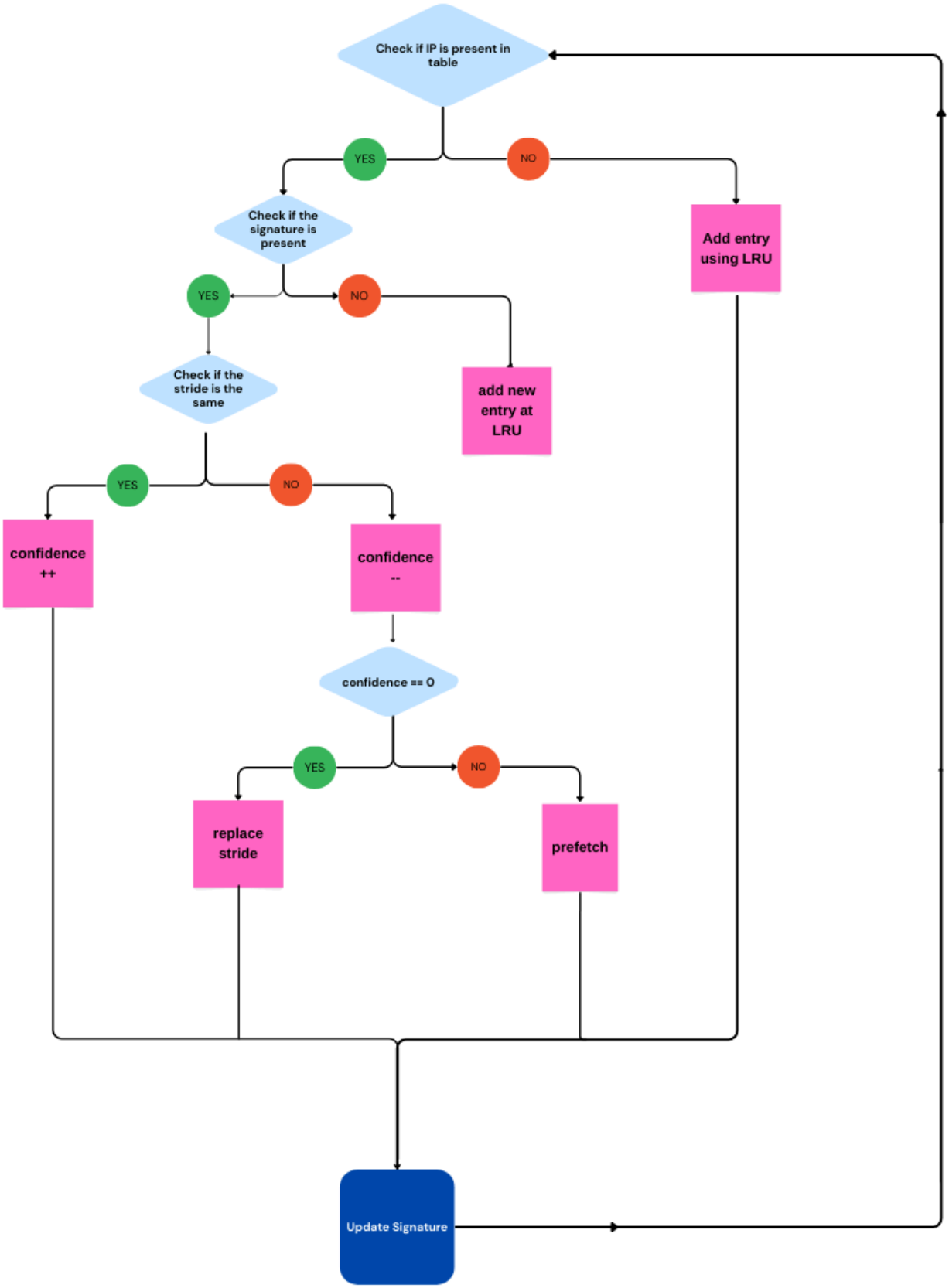
Task 2: Data Prefetcher (5 points)

Metric	baseline	IP stride Prefetcher	Complex stride prefetcher
Instructions	25000000	25000000	25000000
Cycles	255770901	83301596	67557401
IPC	0.0977437	0.300114	0.373718
Speedup vs Baseline	1.000	3.07041	3.823449
L1D Total MPKI	69.9934	70.729	71.239
L1D load MPKI	69.9647	13.7909	4.71292


Speedup Over Baseline Comparison (Higher is Better)
Speedup vs Baseline
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1.0
3.07
3.82
Baseline
IP Stride
Complex Stride
Prefetcher Type


L1D Total & Load MPKI for Different Prefetchers (Lower is Better)
L1D Total MPKI
L1D Load MPKI
MPKI (Lower is Better)
70
60
50
40
30
20
10
0
Baseline
IP Stride
Complex Stride
Prefetcher Type

Complex stride logic flowchart:



Task 3: Guldasta-e-Prefetcher (5 points)

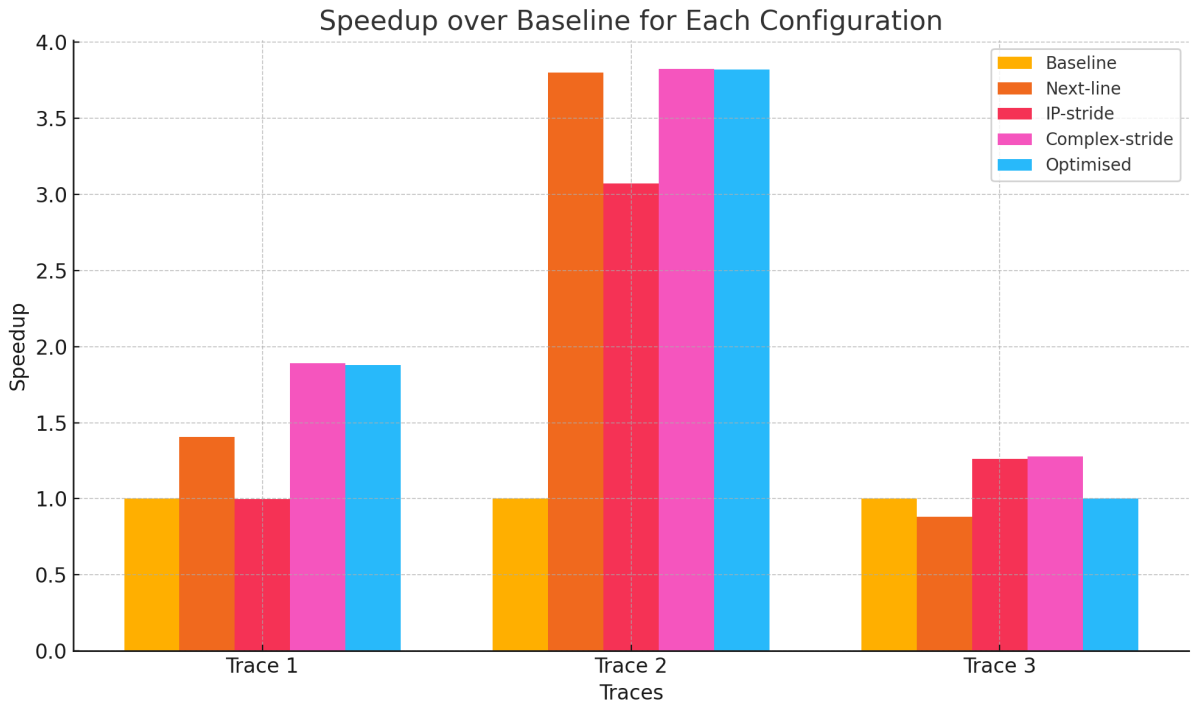
Table for speedup

Prefetcher	Trace 1 Speedup	Trace 2 Speedup	Trace 3 Speedup
Baseline	1.0	1.0	1.0
Next-line	1.4062	3.8006	0.8799
IP-stride	0.9962	3.0704	1.2614
complex-stride	1.8904	3.8234	1.27621
optimised	1.8786	3.8196	1.00012

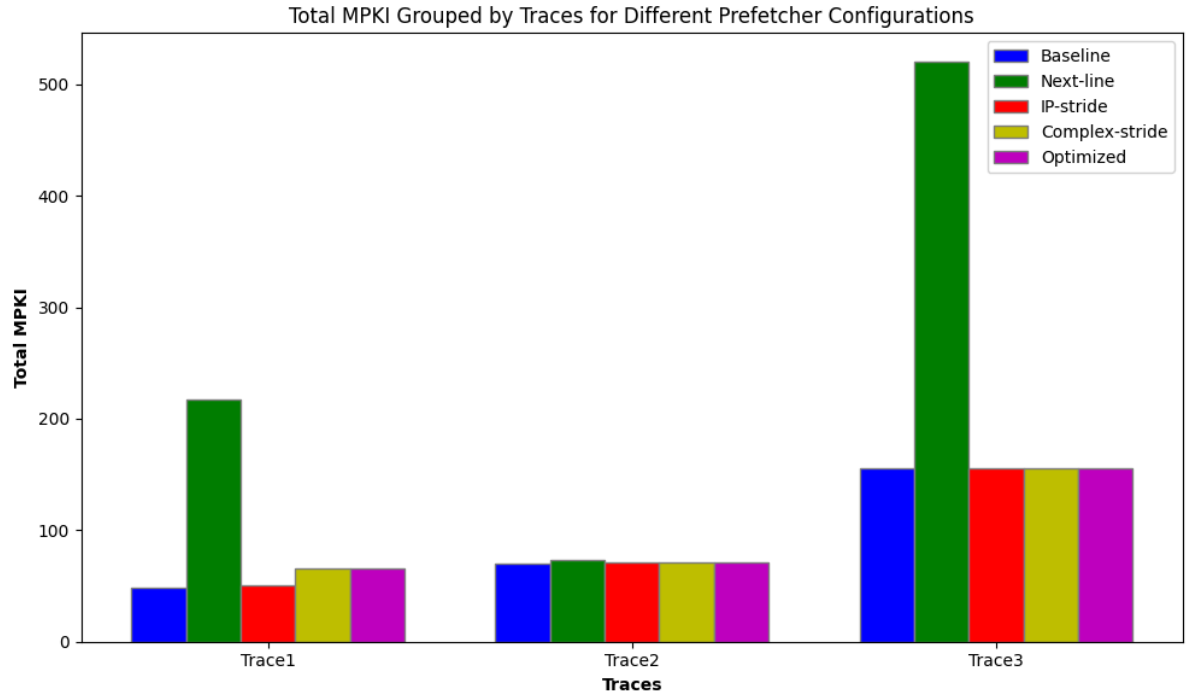
Table for MPKI

Prefetcher	Trace1 Total MPKI	Trace1 Load MPKI	Trace2 Total MPKI	Trace2 Load MPKI	Trace3 Total MPKI	Trace3 Load MPKI
Baseline	48.21	45.4355	69.9934	69.9647	155.279	154.963
Next-line	216.727	49.1029	73.259	11.1521	519.913	121.98
IP-stride	50.7509	45.1975	70.729	13.7909	155.67	112.803
complex-stride	65.4614	37.8805	71.239	4.71292	156.02	112.205
optimised	65.4514	37.954	71.2381	4.757	155.347	154.917

Higher is better



Lower is better



Lower is better:

